哈爾濱工業大學(深圳)
HARBIN INSTITUTE OF TECHNOLOGY, SHENZHEN

Engineering Mechanics

Chapter 3 Spatial Force Systems

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College of Science

3-1 Spatial concurrent force system

3-2 Moment of force about a point and moment of force about an axis

3-3 Spatial force couple

3-4 Equilibrium of arbitrary spatial force system

3-5 The center of gravity of an object

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3-1 Spatial concurrent force system

I、Projection of force on the coordinate axis

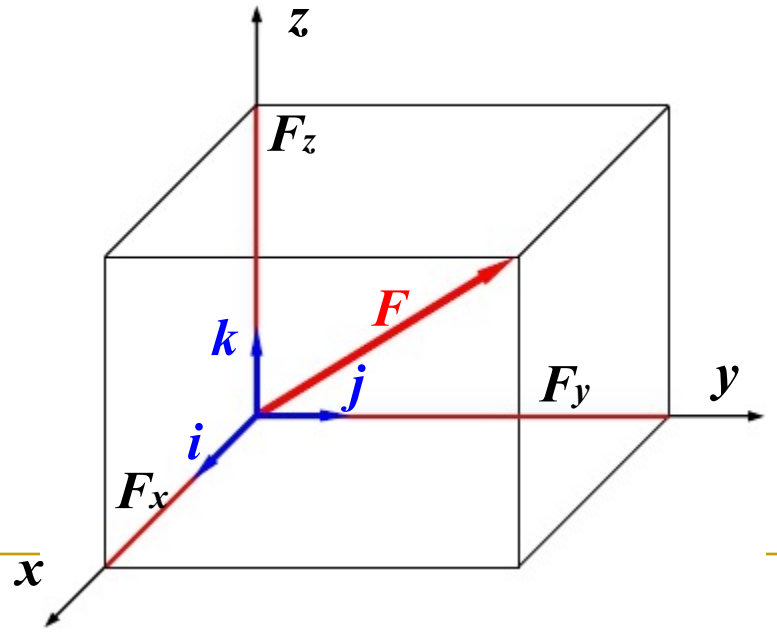
(1) Direct projection method

If the angles between the force and the three axes of the orthogonal coordinate system *Oxyz* are known, the direct projection method is used to represent the projection of force *F* on the three coordinate axes.

$$F_x = F\cos(F,\ i)$$

$$F_y = F\cos(F,\ j)$$

$$F_z = F\cos(F,\ k)$$



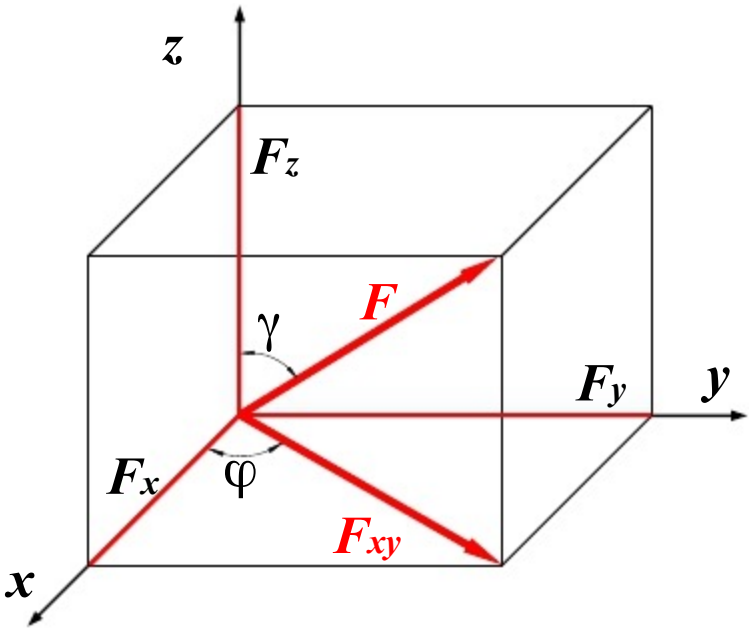
(2) Secondary projection method

When the angle between the force and the coordinate axes Ox 、Oy is difficult to determine, the force F can be projected onto the coordinate plane Oxy to obtain the force Fxy, and then projected onto the x 、y axes. This is called the secondary (indirect) projection method.

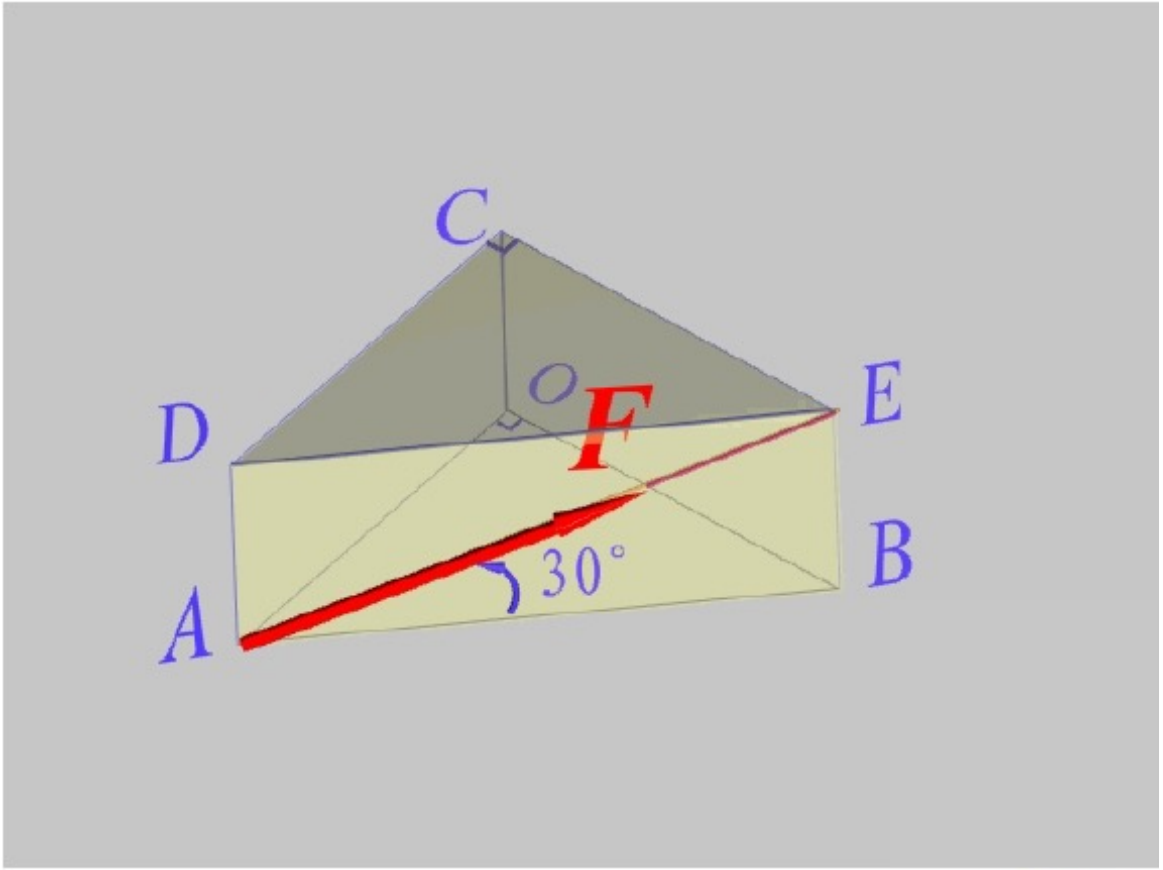
$$F_x = F\sin\gamma\cos\varphi$$

$$F_y = F\sin\gamma\sin\varphi$$

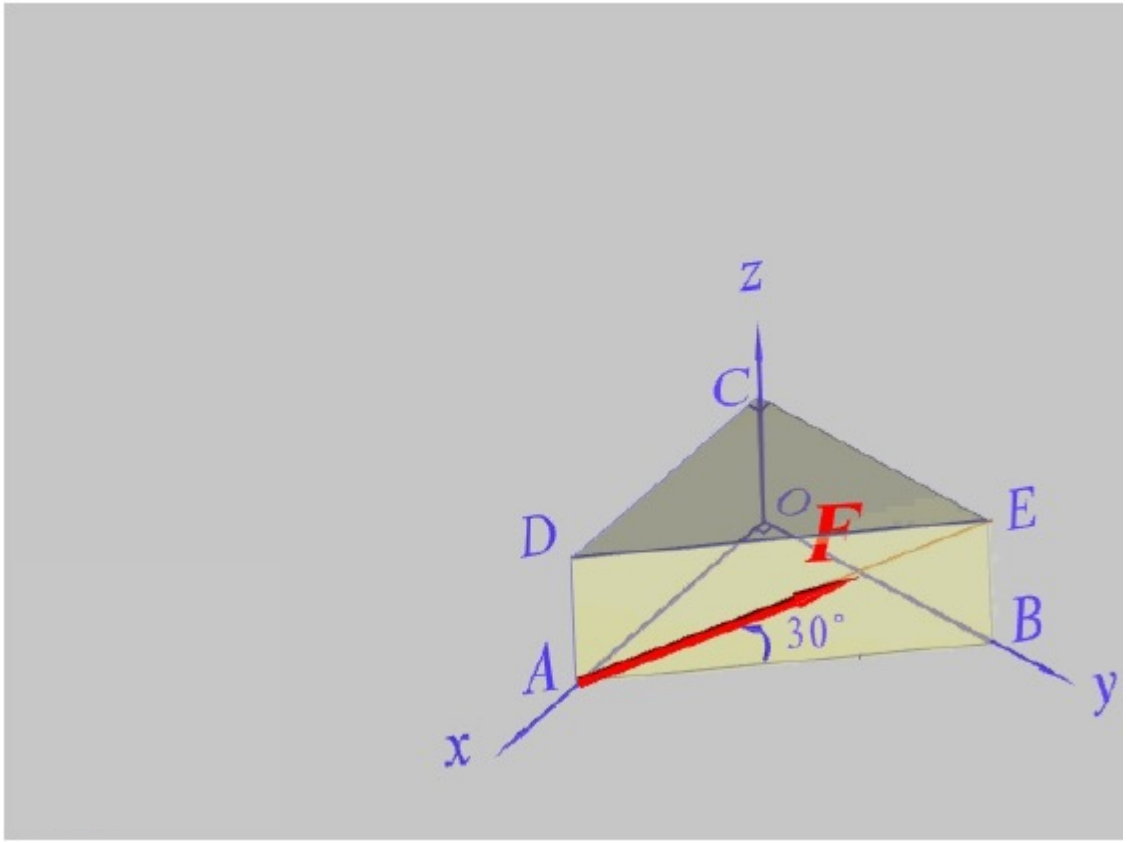
$$F_z = F\cos\gamma$$



Example 1 The bottom surface of a triangular prism is a right angled isosceles triangle, and a force F acts on its side plane $ABED$. The angle between force F and the OAB plane is 30°. Calculate the projection of force F on the three coordinate axes.



Using the quadratic projection method, first project the force *F* onto the Oxy plane, and then project it onto the x, y, z axes respectively.



II. Composition and equilibrium of spatial concurrent forces system

1. Composition： Extending the composition results of planar force systems to spatial force systems

$$\vec{F}_{\mathrm{R}} = \vec{F}_1 + \vec{F}_2 + \cdots + \vec{F}_n = \Sigma \vec{F}_i$$

Or

$$\vec{F}_{\mathrm{R}} = \Sigma F_x \vec{i} + \Sigma F_y \vec{j} + \Sigma F_z \vec{k}$$

The magnitude and direction of the resultant force are：

$$F_{\mathrm{R}} = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2 + (\Sigma F_z)^2}$$

$$\cos(\boldsymbol{F}_{\mathrm{R}}, \ \boldsymbol{i}) = \frac{\Sigma F_x}{F_{\mathrm{R}}}, \ \cos(\boldsymbol{F}_{\mathrm{R}}, \ \boldsymbol{j}) = \frac{\Sigma F_y}{F_{\mathrm{R}}}, \ \cos(\boldsymbol{F}_{\mathrm{R}}, \ \boldsymbol{k}) = \frac{\Sigma F_z}{F_{\mathrm{R}}}$$

2. Equilibrium

The necessary and sufficient condition for the equilibrium of the spatial concurrent force system in space is that the resultant force is equal to zero.

$$\boldsymbol{F}_R = \sum \boldsymbol{F}_i = 0$$

Expressed in analytical form as

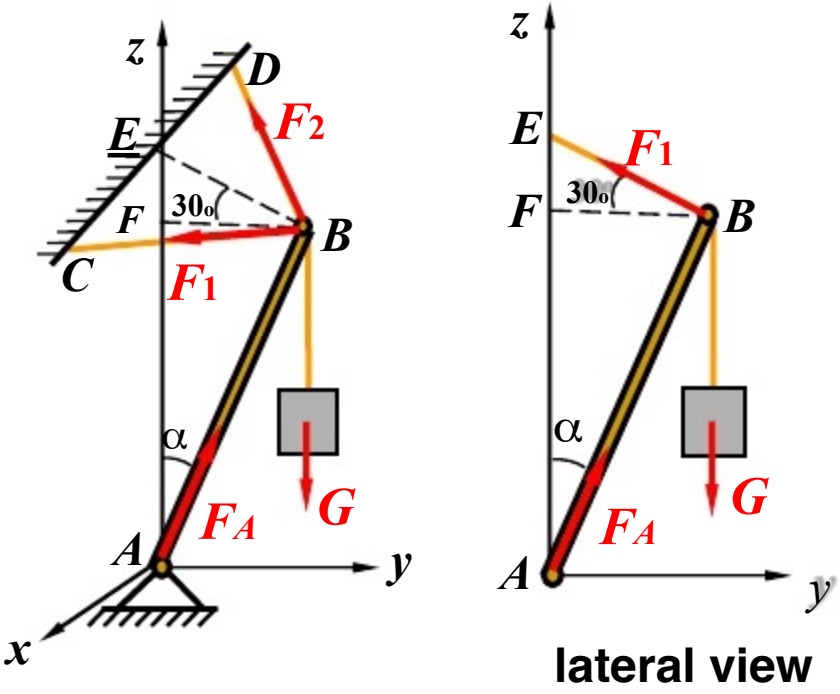
$$\sum F_x = 0$$

$$\sum F_y = 0$$

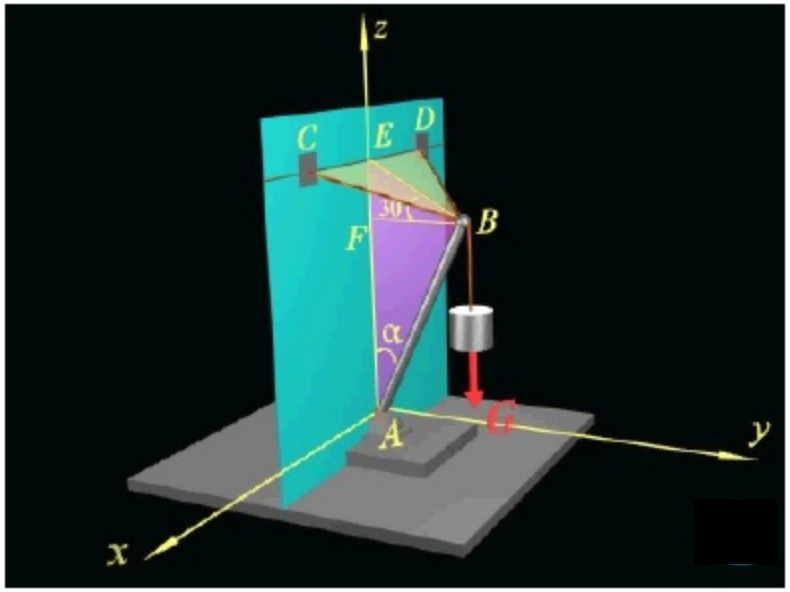
$$\sum F_z = 0$$

The necessary and sufficient condition for the equilibrium of a spatial intersecting force system is that the algebraic sum of the projections of all forces in the force system on the three coordinate axes is equal to zero.

Example 2 The diagram shows the rod *AB* lifting heavy objects. The A end of the lifting rod is fixed to the ground with a smooth ball hinge, and the B end is held by ropes CB and DB, which are respectively tied to points C and D on the wall. The line CD is parallel to the x-axis. known ***CE=EB=DE***，α **=30°**，***The angle between the CDB plane and the horizontal plane*** $\angle EBF$**= 30°**，***G*=10kN**。If the weight of the lifting rod is not considered, try to calculate the force acting on the lifting rod and the tension of the rope.



lateral view



Solution: Take rod AB and a heavy object as the research object, draw a free body force diagram

2. Equilibrium equation

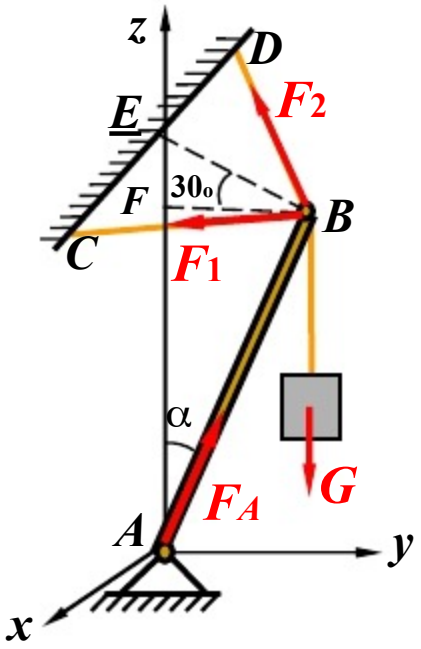
$$\sum F_x = 0 \qquad F_1 \sin 45^\circ - F_2 \sin 45^\circ = 0$$

$$\sum F_y = 0 \qquad F_A \sin 30^\circ - F_1 \cos 45^\circ \cos 30^\circ - F_2 \cos 45^\circ \cos 30^\circ = 0$$

$$\sum F_z = 0 \qquad F_1 \cos 45^\circ \sin 30^\circ + F_2 \cos 45^\circ \sin 30^\circ + F_A \cos 30^\circ - G = 0$$

3. Solve

$$F_1 = F_2 = 3.54\ kN, \quad F_A = 8.66\ kN$$



3-1 Spatial concurrent force system

3-2 Moment of force about a point and moment of force about an axis

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3-4 Equilibrium of arbitrary spatial force system

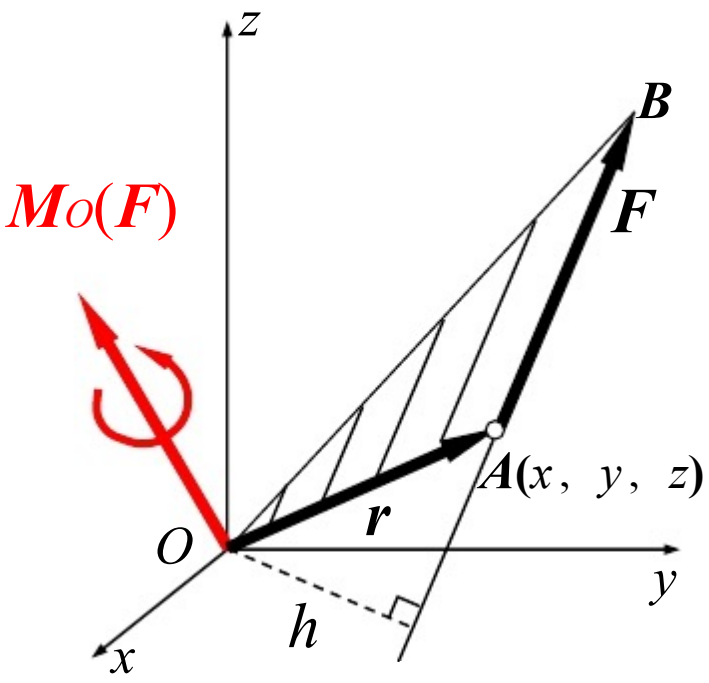
3-5 The center of gravity of an object

3-2 Moment of force about a point and moment of force about an axis

I. The effect of spatial force on the moment about a point depends on：

1. The magnitude of the moment
2. The direction of the moment (sense of rotation)
3. Direction of moment axis

These three factors can be represented by a vector ***Mo*(*F*)**



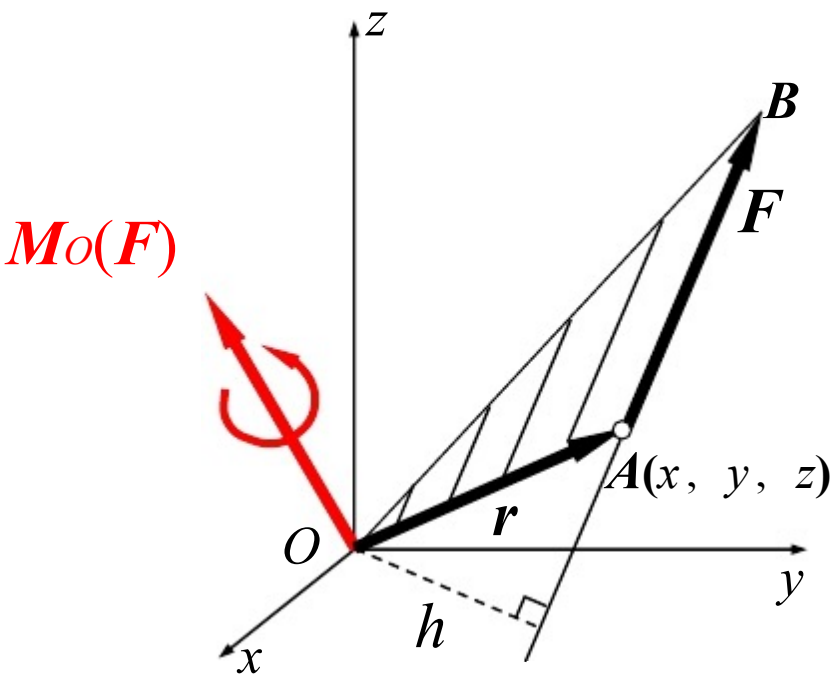
3-2 Moment of force about a point and moment of force about an axis

I. The effect of spatial force on the moment about a point

***Mo*(*F*):**

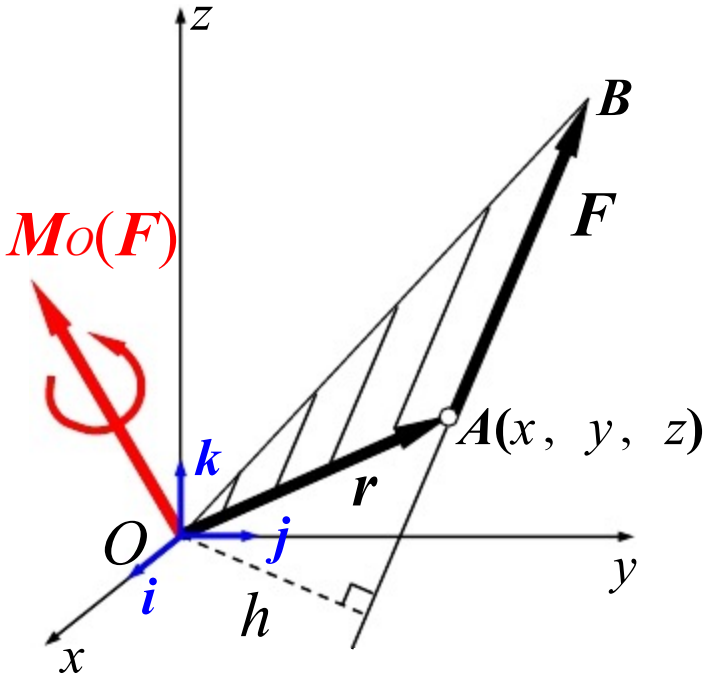
Its modulus represents the magnitude of the moment;

The direction of the moment (sense of rotation)
And
Direction of moment axis are defined by right hand rule



If r represents the radius of force vector acting on point A, then

$$\vec{M}_O(\vec{F}) = \vec{r} \times \vec{F}$$



If a coordinate system is established with the centroid O as the origin, then

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\vec{F} = F_x\vec{i} + F_y\vec{j} + F_z\vec{k}$$

$$\therefore \vec{M}_O(\vec{F}) = \vec{r} \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix}$$

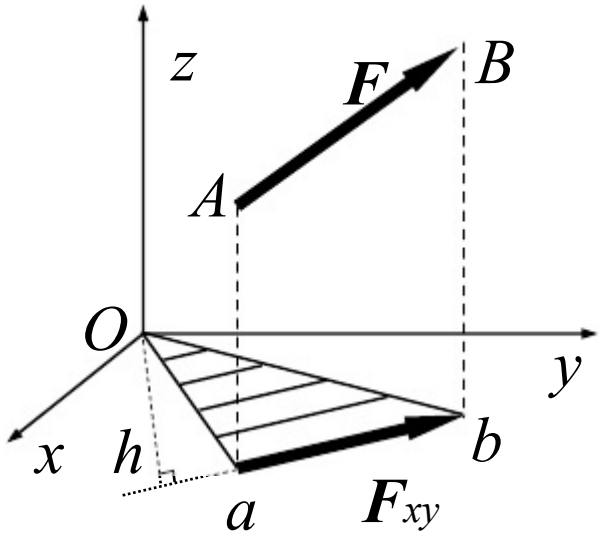
$$= (yF_z - zF_y)\vec{i} + (zF_x - xF_z)\vec{j} + (xF_y - yF_x)\vec{k}$$

$$= [\vec{M}_O(\vec{F})]_x\vec{i} + [\vec{M}_O(\vec{F})]_y\vec{j} + [\vec{M}_O(\vec{F})]_z\vec{k}$$

II、Moment of force about an axis

1、The definition of the moment of force on the axis

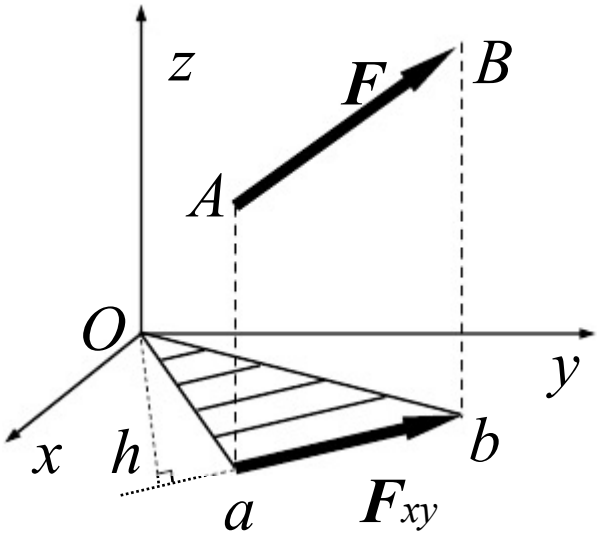
The moment of force about an axis is the force that causes a rigid body to rotate around that axis;
The measure of effectiveness is an algebraic quantity



The absolute value of the moment of force on an axis is equal to the moment of the projection of the force on the plane perpendicular to the axis at its intersection with that plane.

$$M_z(\boldsymbol{F}) = M_O(\boldsymbol{F}_{xy}) = \pm F_{xy}h = \pm 2A_{\Delta Oab}$$

II、Moment of force about an axis



1、The definition of the moment of force on the axis

Regulations on signs：

From the **positive z-axis**, if the force causes the rigid body to rotate counterclockwise,

Take the positive sign, otherwise take the negative sign;

The sign can also be determined according to the right-hand spiral rule.

According to the definition:

(1) When the line of action of force is parallel or intersects (coplanar) with the axis, the moment of force on the axis is equal to zero.

(2) When the force moves along the line of action, its moment to the axis remains unchanged.

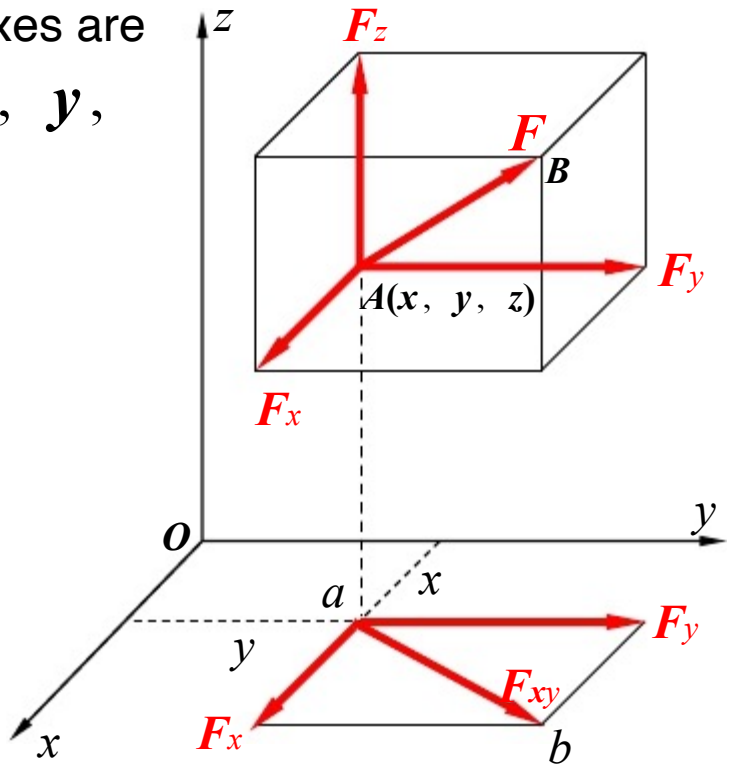
2、Analytical expression for the moment of force about the axis

The projections of force $\boldsymbol{F}$ on the three coordinate axes are F_x, F_y, F_z, the coordinates of force point $\boldsymbol{A}$ are $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z})$, then:

$$M_z(\vec{\boldsymbol{F}}) = Mo(\vec{\boldsymbol{F}_{xy}})$$

$$= Mo(\vec{\boldsymbol{F}_x}) + Mo(\vec{\boldsymbol{F}_y})$$

$$= xF_y - yF_x$$



Similarly, the other two equations:

$$M_x(\vec{\boldsymbol{F}}) = yF_z - zF_y$$

$$M_y(\vec{\boldsymbol{F}}) = zF_x - xF_z$$

$$M_z(\vec{\boldsymbol{F}}) = xF_y - yF_x$$

3、The relationship between the moment of force about a point and the moment of force about the axis passing through that point

The analytical expressions for comparing the moment of force to a point and the moment of force to an axis are as follows:

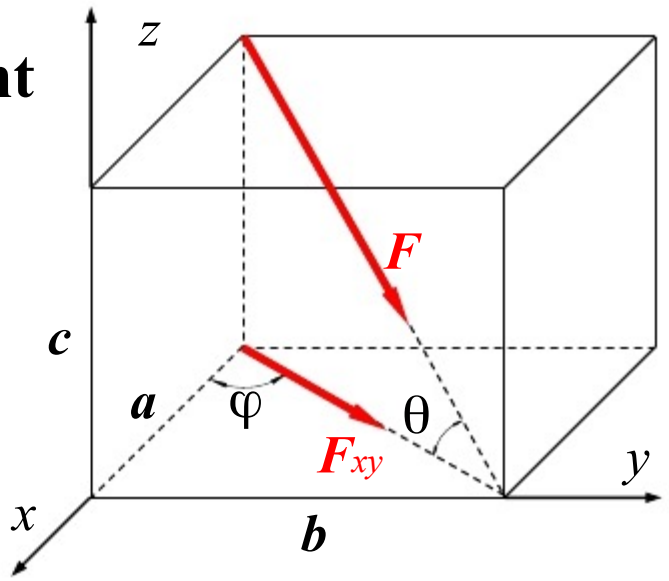
$$[\vec{\boldsymbol{M}}_O(\vec{\boldsymbol{F}})]_x = M_x(\boldsymbol{F})$$

$$[\vec{\boldsymbol{M}}_O(\vec{\boldsymbol{F}})]_y = M_y(\boldsymbol{F})$$

$$[\vec{\boldsymbol{M}}_O(\vec{\boldsymbol{F}})]_z = M_z(\boldsymbol{F})$$

That is, the projection of the moment vector of a spatial force on a certain axis passing through that point is equal to the moment of the force on that axis.

[Example 3] Calculate the projection of force F on the three axes and the moment about the three axes.



$$F_x = F\cos\theta\cos\varphi = \frac{Fa}{\sqrt{a^2+b^2+c^2}}$$

$$F_y = F\cos\theta\sin\varphi = \frac{Fb}{\sqrt{a^2+b^2+c^2}}$$

$$F_z = -F\sin\theta = \frac{-Fc}{\sqrt{a^2+b^2+c^2}}$$

$$\cos\theta = \frac{\sqrt{a^2+b^2}}{\sqrt{a^2+b^2+c^2}}$$

$$\cos\varphi = \frac{a}{\sqrt{a^2+b^2}}$$

$$M_x(F) = M_x(F_x)+M_x(F_y)+M_x(F_z) = -F_y c$$

$$M_y(F) = 0$$

$$M_z(F) = M_z(F_x)+M_z(F_y)+M_z(F_z) = -F_y a$$

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3-3 Spatial force couple

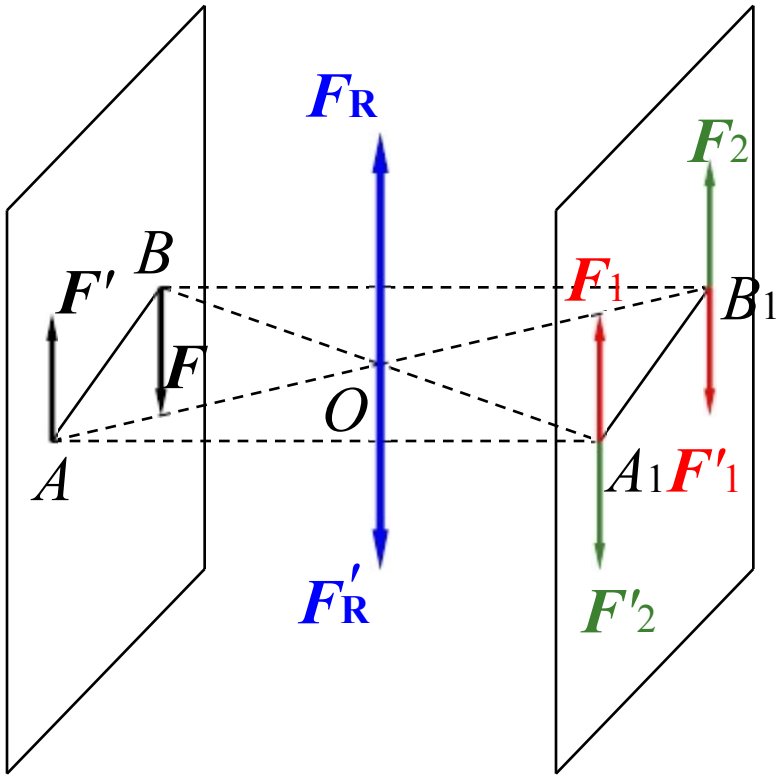
3-4 Equilibrium of arbitrary spatial force system

3-5 The center of gravity of an object

3-3 Spatial force couple

I、Vector representation of force couples

Property: The movement of a couple from one plane parallel to a rigid body to another parallel plane does not affect its effect on the rigid body.



The effectiveness of a force couple depends on the **magnitude** of its moment, **the direction of its rotation,** and the **orientation of moment axis.**

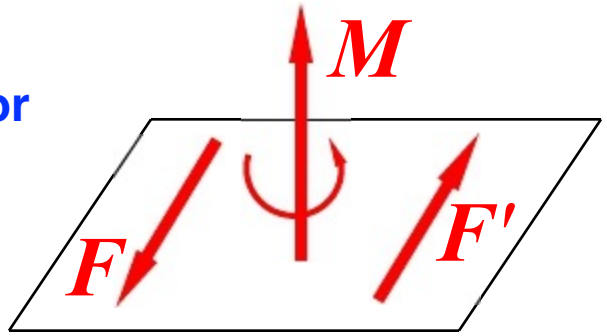
Therefore, a vector $\boldsymbol{M}$ can be used to represent:

the magnitude of the couple moment can be represented by the modulus of $\boldsymbol{M}$;

The direction of M represents the turning of the couple of forces according to the right-hand rule;

The direction of the action line of $\boldsymbol{M}$ is the same as the normal direction of the interaction surface of the force couple.

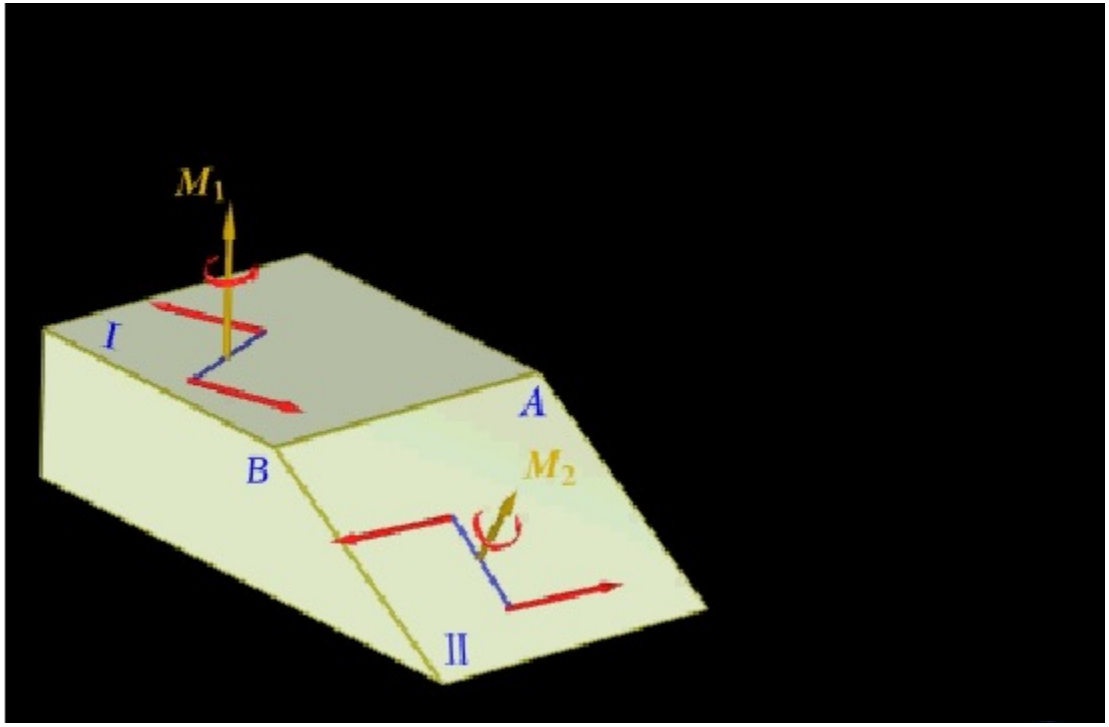
As shown in the figure. $\boldsymbol{M}$ is called the **moment vector** of the force couple.



II、Equivalent Theorem of Space Couple Forces

The equivalent condition for a spatial couple is that two couples acting on the same rigid body are equivalent if their moment vectors are equal.

Composition of spacial force couple systems



III、Composition of spacial force couple systems

1, Composition

$$\vec{M} = \vec{M}_1 + \vec{M}_2 + \cdots + \vec{M}_n = \sum \vec{M}_i$$

According to the theorem of resultant vector projection

$$M_x = \sum M_x \,,\; M_y = \sum M_y \,,\; M_z = \sum M_z$$

The magnitude and direction of the resultant moment are determined by the following equation:

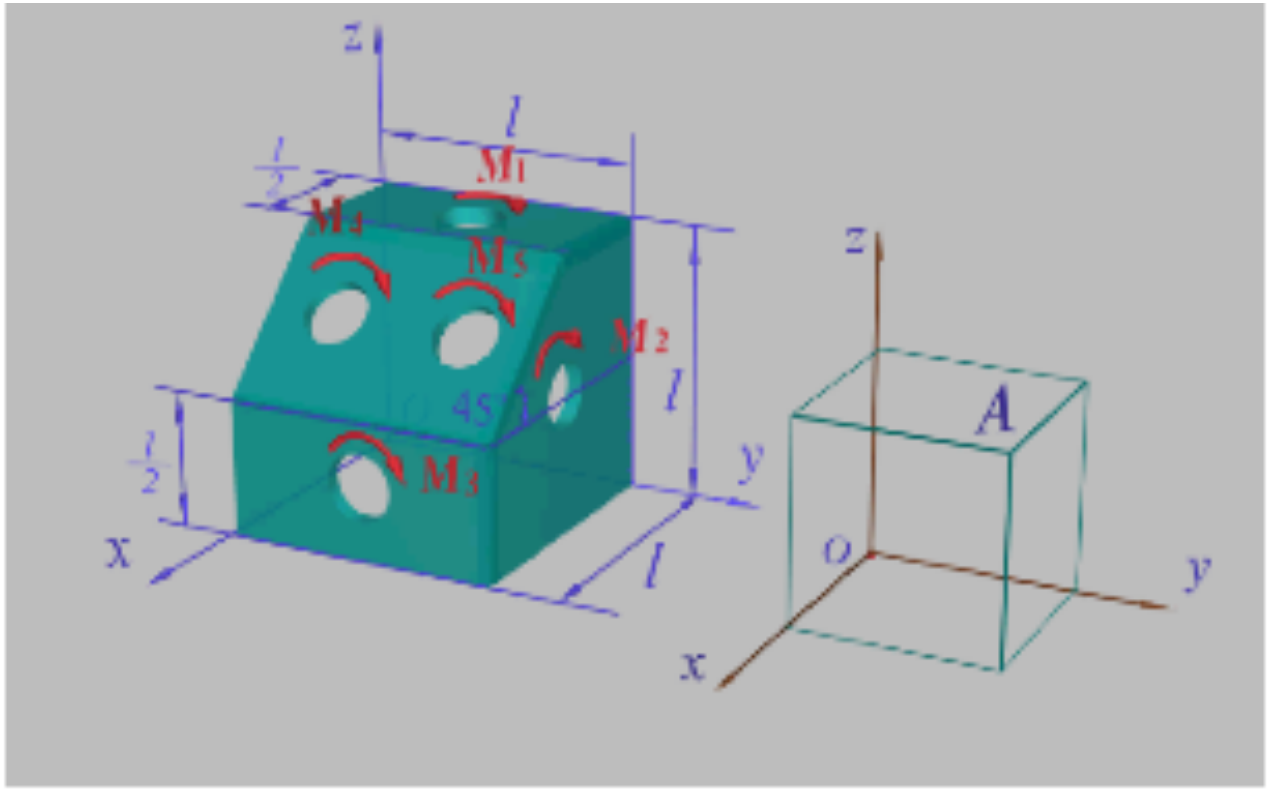
$$M = \sqrt{(\sum M_x)^2 + (\sum M_y)^2 + (\sum M_z)^2}$$

$$\cos(\boldsymbol{M},\ \boldsymbol{i}) = \frac{M_x}{M}$$

$$\cos(\boldsymbol{M},\ \boldsymbol{j}) = \frac{M_y}{M}$$

$$\cos(\boldsymbol{M},\ \boldsymbol{k}) = \frac{M_z}{M}$$

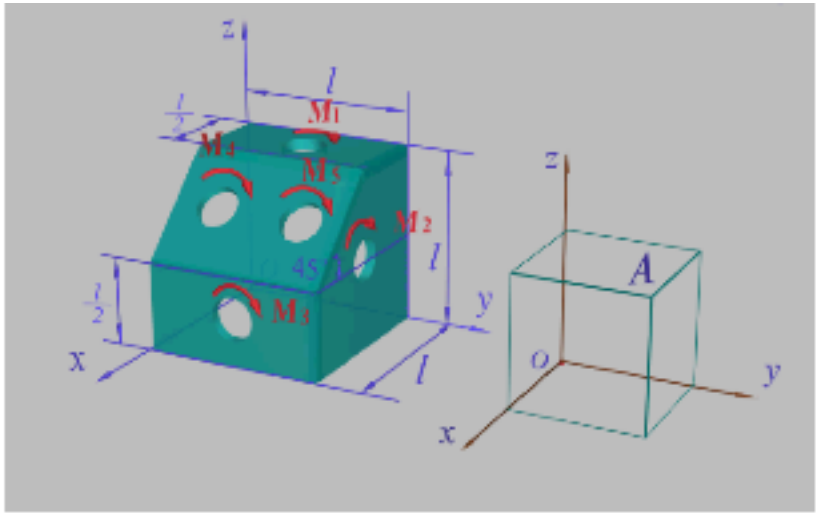
Example 4: As shown in the figure, five holes are simultaneously drilled on the four faces of the workpiece, with each hole subjected to a cutting force couple moment of ***80 N·m***. Determine the projections of the **resultant couple** moment on the x, y, and z axes (***M_x, M_y, M_z***), and calculate the magnitude and direction of the resultant couple moment vector.



Solution:

Composition of spacial force couple system

$$M_x = -M_3 - M_4\cos 45^\circ - M_5\cos 45^\circ = -193.1\,\mathrm{N\cdot m}$$
$$M_y = -M_2 = -80\,\mathrm{N\cdot m}$$
$$M_z = -M_1 - M_4\cos 45^\circ - M_5\cos 45^\circ = -193.1\ \ \mathrm{N\cdot m}$$



The magnitude of resultant moment vector:

$$M = \sqrt{M_x^2 + M_y^2 + M_z^2} = 284.6\ \mathrm{N\cdot m}$$

The cosine direction of the resultant moment vector:

$$\cos(\boldsymbol{M},\ \boldsymbol{i}) = -0.6786$$
$$\cos(\boldsymbol{M},\ \boldsymbol{j}) = -0.2811$$
$$\cos(\boldsymbol{M},\ \boldsymbol{k}) = -0.6786$$

2, Equilibrium: The spatial force couple system can compose a resultant force couple, so the necessary and sufficient condition for the equilibrium of the spatial force couple system is that the moment vector of the resultant force couple is equal to zero. Namely:

$$\vec{M} = \vec{M}_1 + \vec{M}_2 + \cdots + \vec{M}_n = \sum \vec{M}_i = 0$$

Because: $$M = \sqrt{(\sum M_x)^2 + (\sum M_y)^2 + (\sum M_z)^2}$$

So:

$$\begin{cases} \sum M_x = 0 \\ \sum M_y = 0 \\ \sum M_z = 0 \end{cases}$$

The above equations are the equilibrium equations of the spatial couple system

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3-4 Equilibrium of arbitrary spatial force system

3-5 The center of gravity of an object

3-4 Equilibrium of arbitrary spatial force system

I、The equilibrium equations of arbitrary spacial force system are:

$\boldsymbol{F}'_R=0$, $\boldsymbol{M}_O=0$, known as:

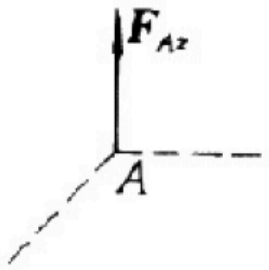
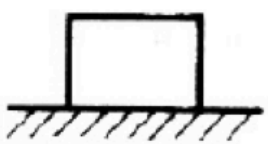
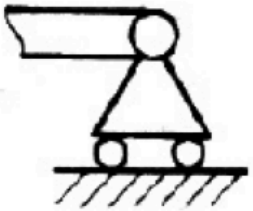
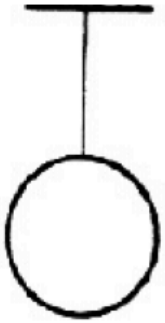
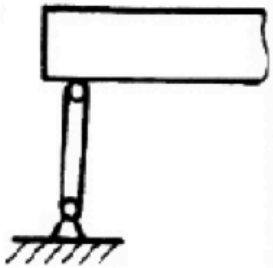
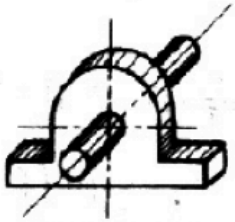
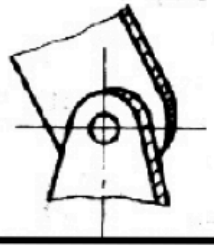
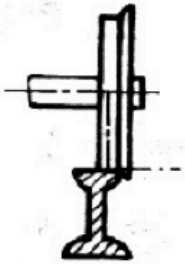
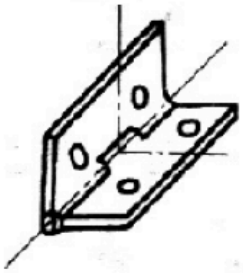
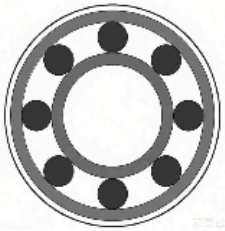
$$\sum F_x = 0, \quad \sum F_y = 0, \quad \sum F_z = 0$$

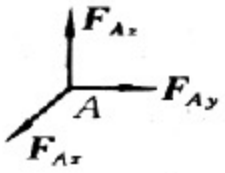
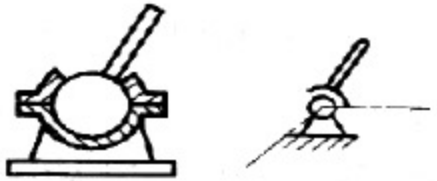
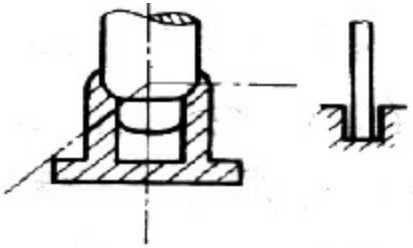
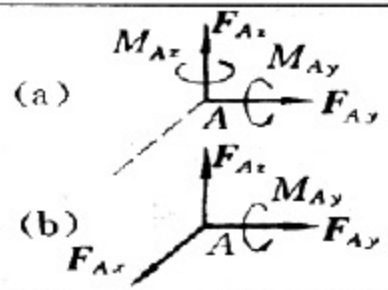
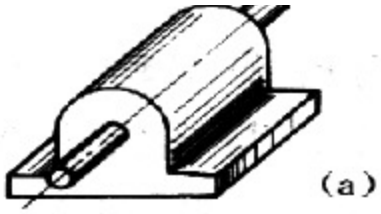
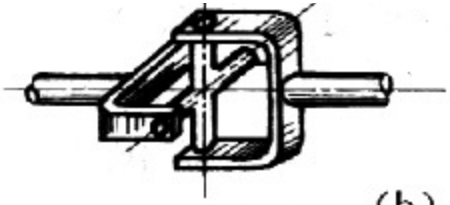
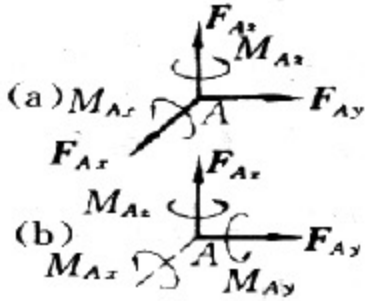
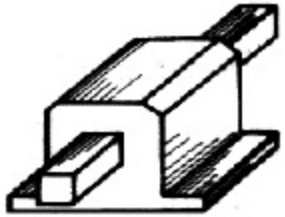
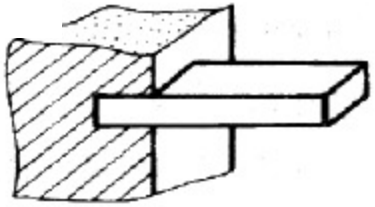
$$\sum M_x(\boldsymbol{F}) = 0, \quad \sum M_y(\boldsymbol{F}) = 0, \quad \sum M_z(\boldsymbol{F}) = 0$$

The **necessary and sufficient conditions** for the equilibrium of arbitrary force system in space are that the algebraic sum of the **projections** of each force on the three coordinate axes in the force system is equal to zero, and the algebraic sum of the **moments** of each force on the three axes is also equal to zero.

The above equations are the equilibrium equations of arbitrary force system in space.

II、 Space constraint types

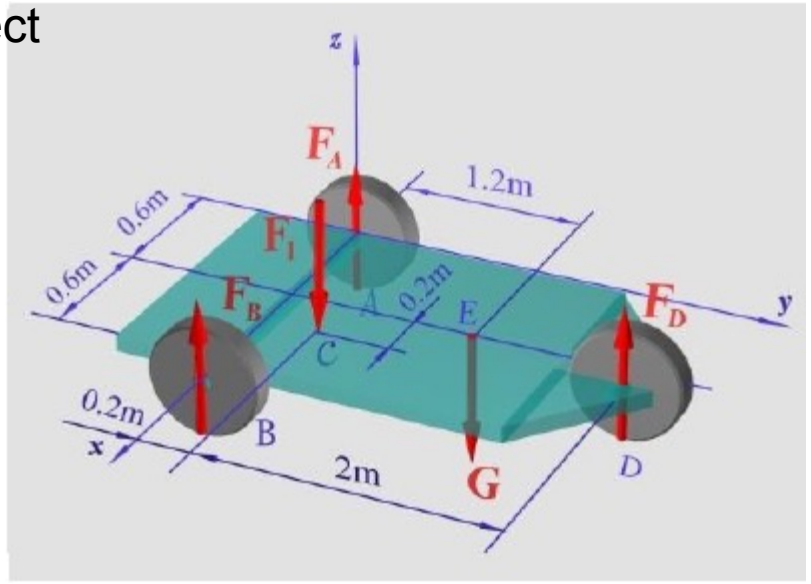
	Constrained forces	Constrained types
1		Smooth surface  Flexible hinge Support  Non extensible cable  
2		Guide bearing  Clamping plate with pin  Rail  Butt hinge  Ball bearing 

3		Ball hinge  Thrust bearing 
4		Guide bearing  (a) Universal joint / coupling  (b)
5		Clamping plate with long pin  (a) Internal Rail  (b)
6		Fixed end 

Example 5: cart with three wheels, weight G=8kN, exert on point E, load F_1=10 kN, exert on point C.

Request the constraint force of the ground on the wheels when the car is stationary.

Solution: Taking the car as the research object



equilibrium equations:

$$\sum F_z = 0$$

$$-F_1 - G + F_A + F_B + F_D = 0$$

$$\sum M_x(\boldsymbol{F}) = 0$$

$$-F_1 \cdot 0.2\text{m} - G \cdot 1.2\text{m} + F_D \cdot 2\text{m} = 0$$

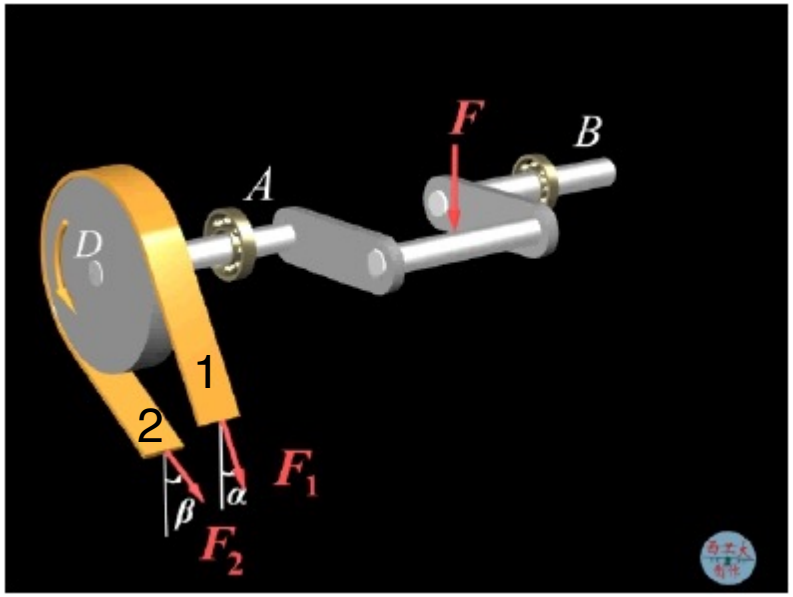
$$\sum M_y(\boldsymbol{F}) = 0$$

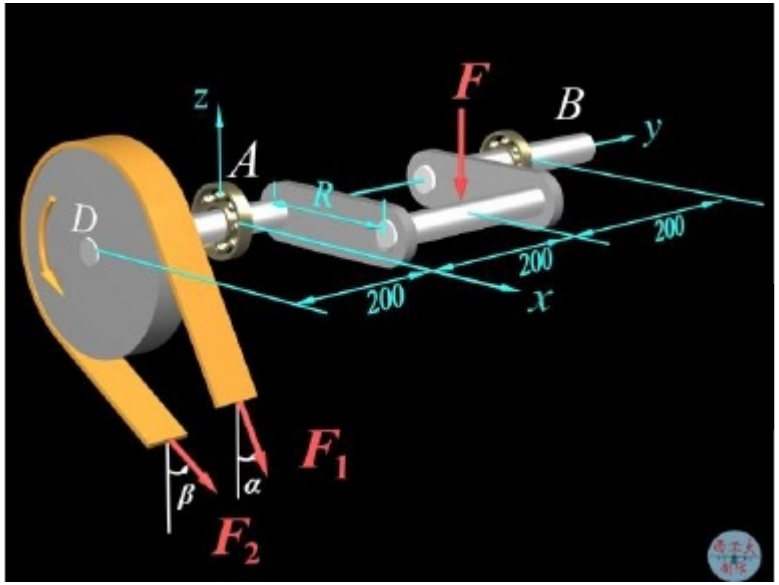
$$F_1 \cdot 0.8\text{m} + G \cdot 0.6\text{m} - F_D \cdot 0.6\text{m} - F_B \cdot 1.2\text{m} = 0$$

Solve: $F_D = 5.8\ kN$, $F_B = 7.777\ kN$, $F_A = 4.423\ kN$

Example 6: The tension of the tape in the figure $F_2=2F_1$, a vertical force acting on the crankshaft F=2kN。the diameter of the tape wheel is known as D=400mm, Crank length R=300mm, The angles between tape end 1 and tape end 2 and the perpendicular line are α and β respectively, $\alpha=30^0$, $\beta=60^0$, other dimensions as shown in the figure.

Calculate the tape tensions and bearing constraint forces





Solution: Taking the axis as the research object, the force diagram is shown:

$$\sum F_x = 0 \quad F_1 \sin 30^\circ + F_2 \sin 60^\circ + F_{Ax} + F_{Bx} = 0$$

$$\sum F_z = 0 \quad -F_1 \cos 30^\circ - F_2 \cos 60^\circ - F + F_{Az} + F_{Bz} = 0$$

$$\sum M_x(\boldsymbol{F}) = 0 \quad F_1 \cos 30^\circ \times 0.2\text{m} + F_2 \cos 60^\circ \times 0.2\text{m} - F \times 0.2\text{m} + F_{Bz} \times 0.4\text{m} = 0$$

$$\sum M_y(\boldsymbol{F}) = 0 \quad FR - \frac{D}{2}(F_2 - F_1) = 0$$

$$\sum M_z(\boldsymbol{F}) = 0 \quad F_1 \sin 30^\circ \times 0.2\text{m} + F_2 \sin 60^\circ \times 0.2\text{m} - F_{Bx} \times 0.4\text{m} = 0$$

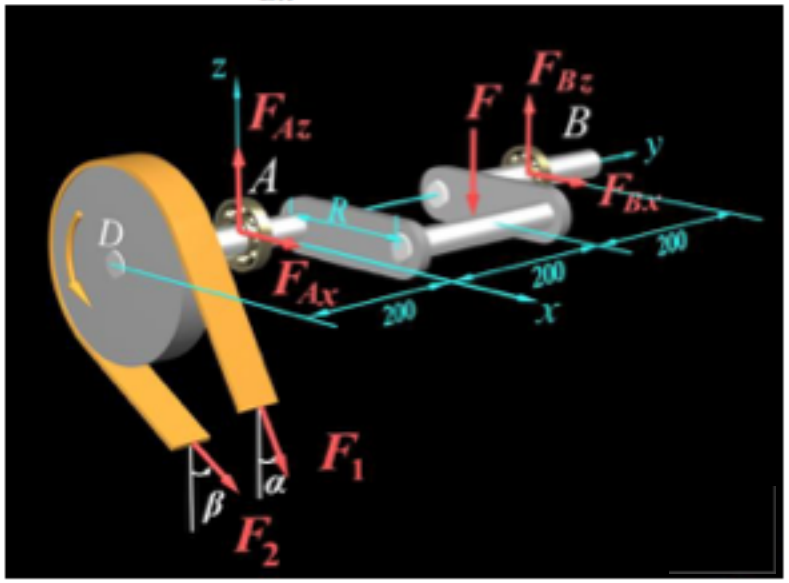
$$F_2 = 2F_1$$

Solve:

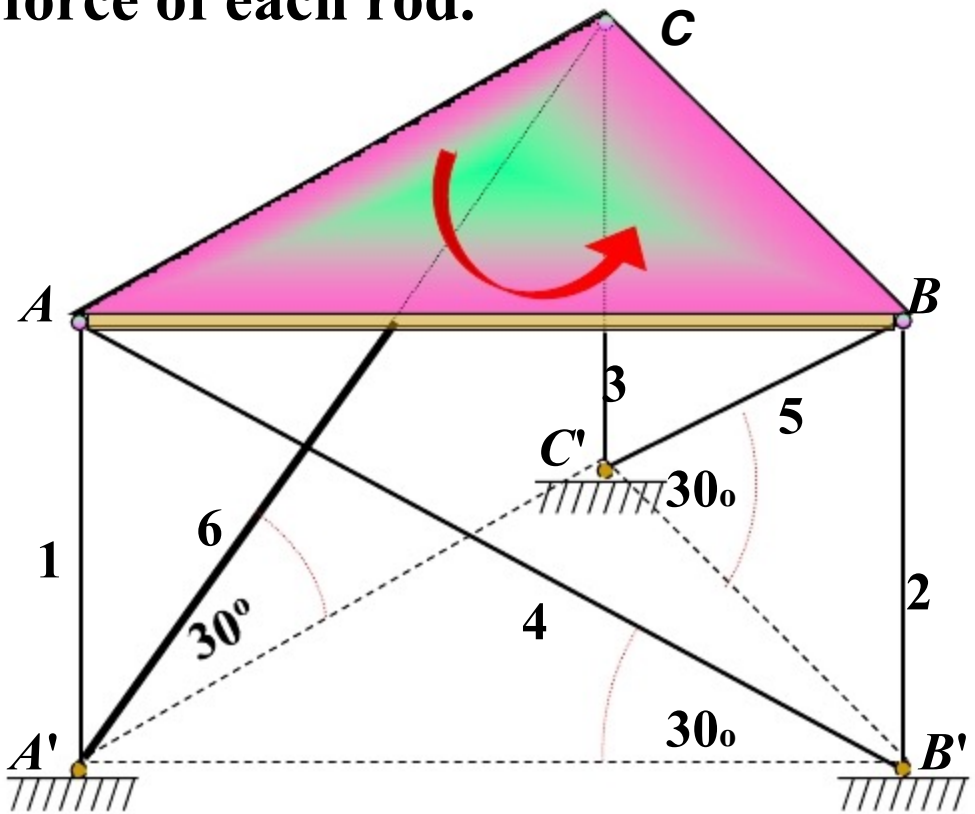
$$F_1 = 3\text{kN}, \quad F_2 = 6\text{kN}$$

$$F_{Ax} = -10.04\text{kN}, \quad F_{Az} = 9.40\text{kN}$$

$$F_{Bx} = 3.35\text{kN}, \quad F_{Bz} = -1.80\text{kN}$$



Example 7: An equilateral triangular plate with a side length of a is supported in a horizontal position by six rods as shown in the figure. A moment is *M* is exerted inside the plate. Calculate the constraint force of each rod.



3-1 Spatial concurrent force system

3-2 Moment of force about a point and moment of force about an axis

3-3 Spatial force couple

3-4 Equilibrium of arbitrary spatial force system

3-5 The center of gravity of an object

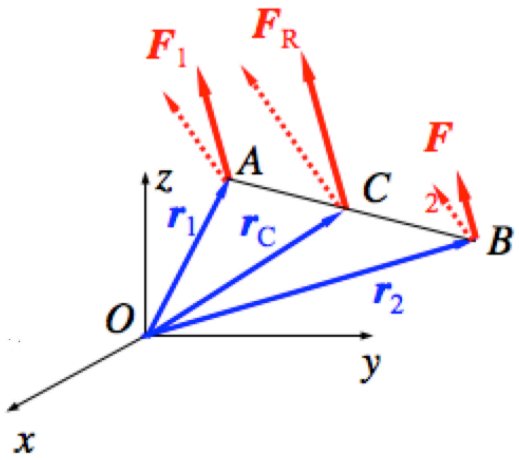
3-5 The center of gravity of an object

1. Center of Parallel Force System

The center of a parallel force system is a point through which the resultant force of the parallel force system passes.

The position of the resultant force is only related to the magnitude of each parallel force and the position of the point of action, and is independent of the direction of each parallel force.

This point is called the center of the parallel force system.



$$\vec{r}_C = \frac{\Sigma F_i \vec{r}_i}{\Sigma F_i} \qquad \left(\vec{r}_C \times F_R \vec{F}^0 = \vec{r}_1 \times F_1 \vec{F}^0 + \cdots + \vec{r}_n \times F_n \vec{F}^0\right)$$

$$x_C = \frac{\Sigma F_i x_i}{\Sigma F_i}, \quad y_C = \frac{\Sigma F_i y_i}{\Sigma F_i}, \quad z_C = \frac{\Sigma F_i z_i}{\Sigma F_i}$$

2. Center of gravity of an object

Gravity is the gravitational force exerted by the Earth on an object.
If an object is viewed as composed of countless particles, then gravity forms the concurrent force system in space.
Due to the much smaller size of objects compared to Earth, it can be approximated that gravity is a parallel force system, and the resultant force of this force system is the weight of the object.
No matter how an object is placed, the line of action of its gravitational force always passes through a certain point relative to the object, which is called the center of gravity of the object.

The formulas for the center of gravity of a **stationary** object are:

$$x_C = \frac{\sum P_i x_i}{\sum P_i}, \quad y_C = \frac{\sum P_i y_i}{\sum P_i}, \quad z_C = \frac{\sum P_i z_i}{\sum P_i}$$

For **homogeneous** objects, homogeneous rods or homogeneous plates, their center of gravity coordinates are respectively:

$$x_C = \frac{\int_V x\mathrm{d}V}{V}, y_C = \frac{\int_V y\mathrm{d}V}{V}, z_C = \frac{\int_V z\mathrm{d}V}{V}$$

$$x_C = \frac{\int_A x\mathrm{d}A}{A}, y_C = \frac{\int_A y\mathrm{d}A}{A}, z_C = \frac{\int_A z\mathrm{d}A}{A}$$

$$x_C = \frac{\int_l x\mathrm{d}l}{l}, y_C = \frac{\int_l y\mathrm{d}l}{l}, z_C = \frac{\int_l z\mathrm{d}l}{l}$$

The center of gravity of a homogeneous object is its geometric center, also known as the **centroid**.

3. Method for determining the center of gravity of an object

(1) The center of gravity of a simple geometric object

If a homogeneous object has a symmetry plane, symmetry axis, or symmetry center, then the center of gravity of the object must correspondingly lie on this symmetry plane, symmetry axis, or symmetry center.

The center of gravity of a simple shaped object can be found in the Engineering Manual.

(2) Find the center of gravity using the combination method

① Segmentation method

If an object is composed of several simple shaped objects, and the center of gravity of these objects is known, then the center of gravity of the entire object can be calculated by the following equation:

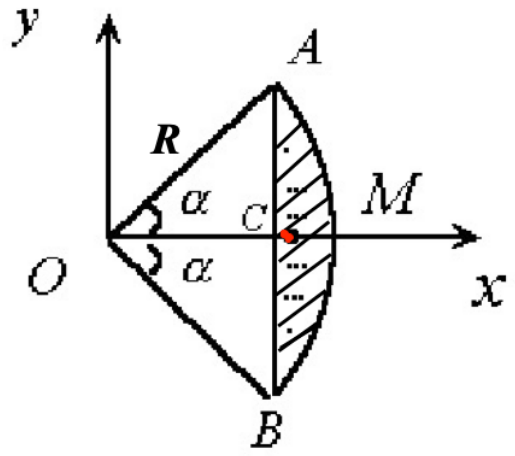
$$x_C = \frac{\sum P_i x_i}{\sum P_i}, \quad y_C = \frac{\sum P_i y_i}{\sum P_i}, \quad z_C = \frac{\sum P_i z_i}{\sum P_i}$$

② Negative area method

If a part of an object or thin plate is cut off (such as an object with holes or voids), the center of gravity of such object can still be calculated using the same formula as the segmentation method, except that the volume or area of the cut off part should be taken as a negative value.

Example 7: The diagram shows a homogeneous and equally thick block, whose cross-sectional area is enclosed by an arc *AMB* with a radius of *R* and a line segment *AB* forming an arch shape. Please find the position of the center of gravity of the bow formed by AB in its symmetry plane.

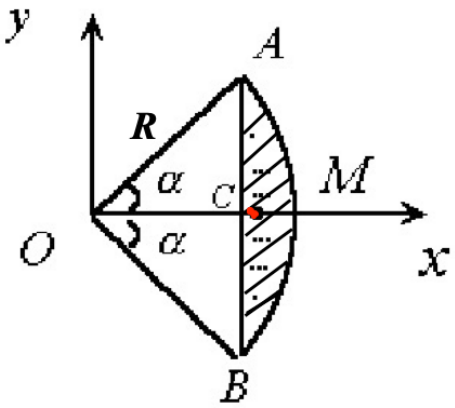
Solution **1**、Establish a Cartesian coordinate system ***oxy*** on the symmetry plane of the object block. From the symmetry: $y_c=0$。



2、The area of the bow shown in the diagram can be obtained by removing the triangle *OAB* from the sector *OAMB*, and the center of gravity of the bow can be determined by the **negative area method**. The area and center of gravity of the fan-shaped and triangular rows can be obtained by referring to the engineering manual;

Sectorial area $OAMB$: $A_1 = R^2\alpha$

Formula for center of gravity position: $x_1 = \frac{2}{3}\frac{R\sin\alpha}{\alpha}$



The area of triangle OAB:

$$A_2 = -\frac{1}{2}(2R\sin\alpha)(R\cos\alpha) = -R^2\sin\alpha\cos\alpha$$

Formula for center of gravity position: $x_2 = \frac{2}{3}(R\cos\alpha)$

$$x_c = \frac{A_1x_1 + A_2x_2}{A_1 + A_2} = \frac{\frac{2}{3}R^3\sin\alpha - \frac{2}{3}R^3\sin\alpha\cos^2\alpha}{R^2\alpha - R^2\sin\alpha\cos\alpha}$$

$$= \frac{2R\sin\alpha(1-\cos^2\alpha)}{3(\alpha-\sin\alpha\cos\alpha)} = \frac{4R\sin^3\alpha}{3(2\alpha-\sin 2\alpha)}$$

Example 8: Calculate the centroid coordinates of a homogeneous thin plate with equal thickness dimensions as shown in the figure.

Solution: Divided by dashed lines as shown in the figure, there are three small rectangles with areas and coordinates of:

$$x_1 = -15\text{mm},\quad y_1 = 45\text{mm},\quad A_1 = 300\text{mm}^2$$

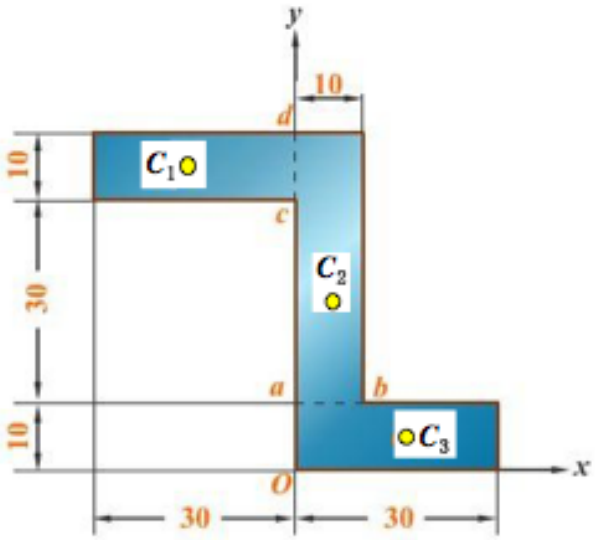
$$x_2 = 5\text{mm},\quad y_2 = 30\text{mm},\quad A_2 = 400\text{mm}^2$$

$$x_3 = 15\text{mm},\quad y_3 = -5\text{mm},\quad A_3 = 300\text{mm}^2$$

Solve:

$$x_C = \sum \frac{A_i x_i}{A} = \frac{A_1 x_1 + A_2 x_2 + A_3 x_3}{A_1 + A_2 + A_3} = 2\text{mm}$$

$$y_C = \sum \frac{A_i y_i}{A} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3} = 27\text{mm}$$



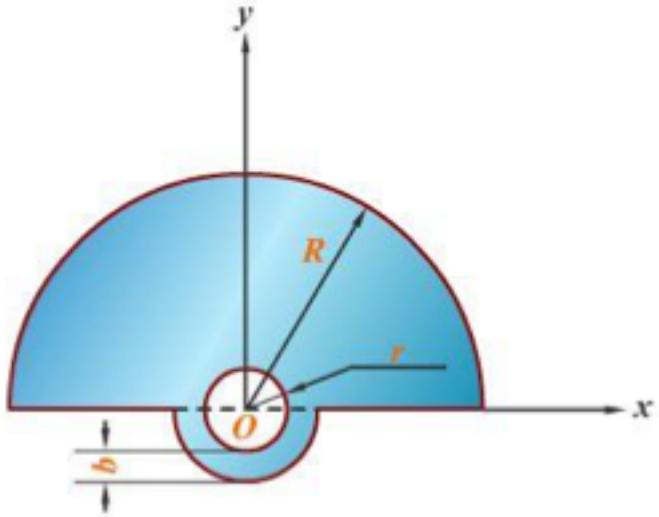
Example 9: Known homogeneous eccentric blocks of equal thickness, *R = 100mm, r = 17mm, b = 13mm*. calculate its center of gravity

Solution: Using the negative area method, it consists of three parts. Let the area of the large semicircle be A_1, the area of the small semicircle (radius r) be A_2, and the area of the small circle A_3

By symmetry $x_C = 0$

$$A_1 = \frac{\pi}{2}R^2,\quad A_2 = \frac{\pi}{2}(r+b)^2,\quad A_3 = -\pi r^2$$

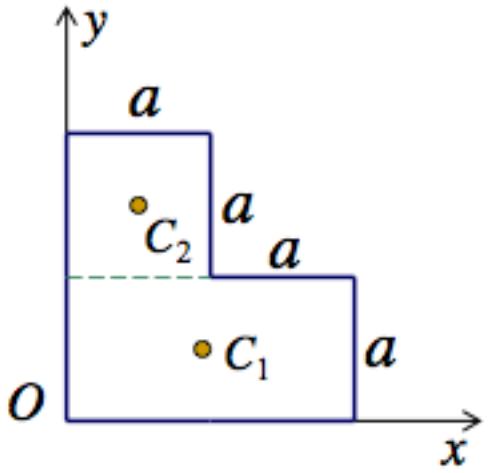
$$y_1 = \frac{4R}{3\pi},\quad y_2 = -\frac{4(r+b)}{3\pi},\quad y_3 = 0$$



From $y_C = \dfrac{\sum A_i y_i}{\sum A}$

Solve: $y_C = \dfrac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3} = 40.01\text{mm}$

Example 10: Calculate the coordinates of the center of gravity of a homogeneous plate as shown in the diagram

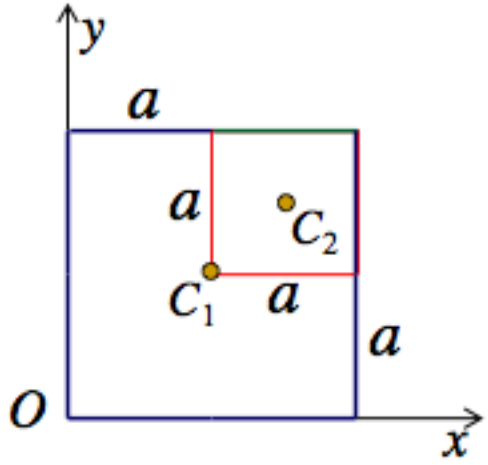


Solution 1: Segmentation Method

$$x_C = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} = \frac{2a^2 a + a^2 \frac{1}{2} a}{3a^2} = \frac{5}{6} a$$

$$y_C = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = \frac{2a^2 \frac{1}{2} a + a^2 \frac{3}{2} a}{3a^2} = \frac{5}{6} a$$

Solution 2: Negative Area Method

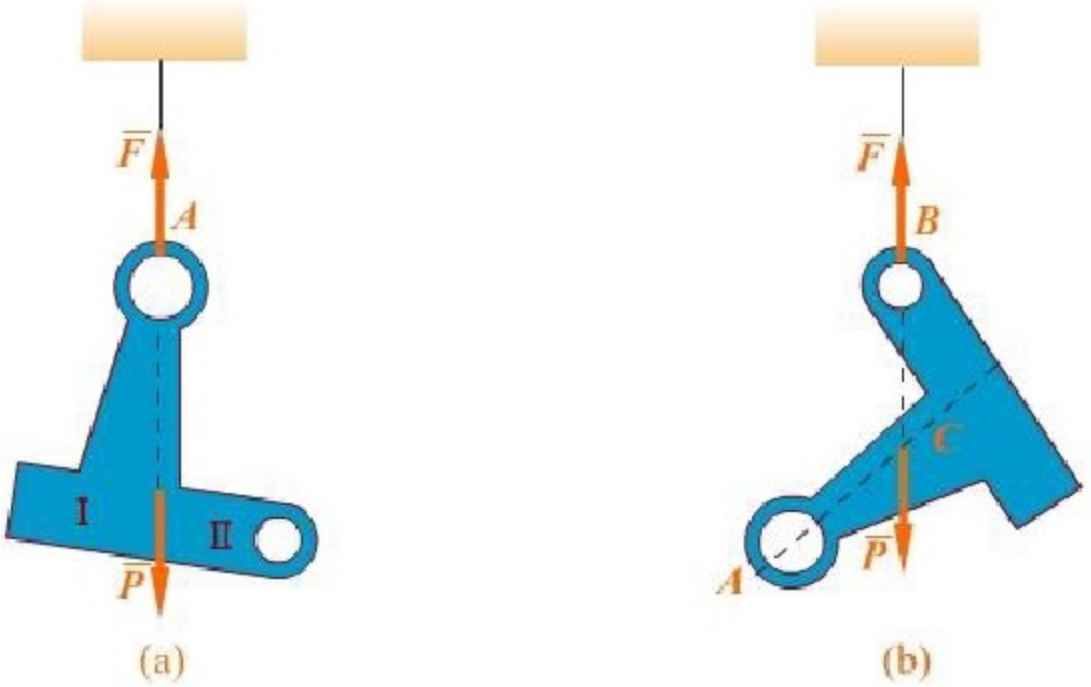


$$x_C = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} = \frac{4a^2 a + (-a^2) \frac{3}{2} a}{4a^2 + (-a^2)} = \frac{5}{6} a$$

$$y_C = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = \frac{4a^2 a + (-a^2) \frac{3}{2} a}{4a^2 + (-a^2)} = \frac{5}{6} a$$

（3）Determine the position of the center of gravity through experimental methods

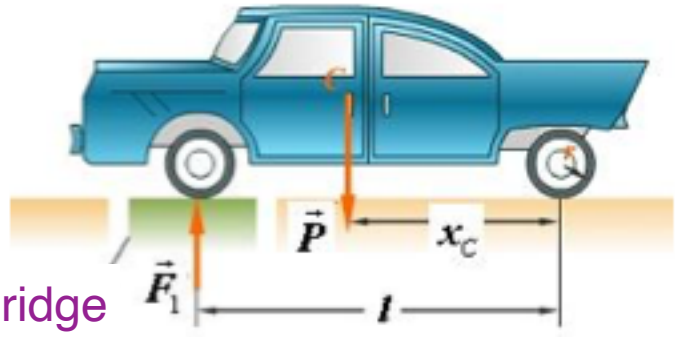
① Hanging method



② weighing method

$$P \cdot x_C = F_1 \cdot l \quad \text{Then} \quad x_C = \frac{F_1}{P} l$$

Rear wheel raised $x_C' = \frac{F_2}{P} l'$

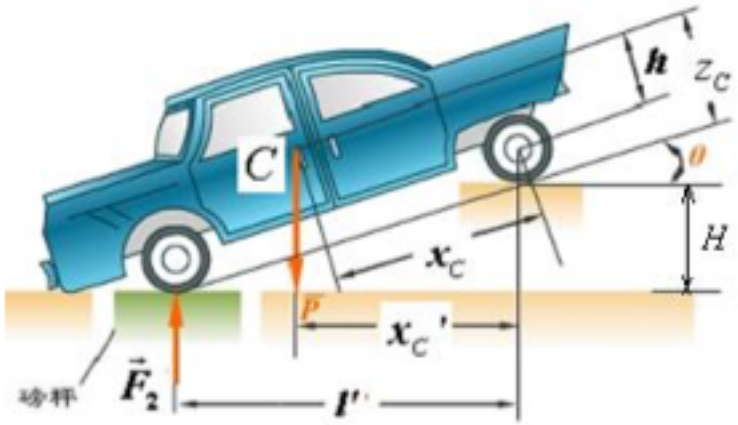


weighbridge

Geometric relationships in the figure $l' = l\cos\theta$

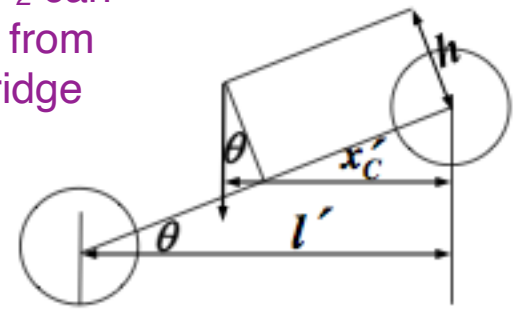
$$x_C' = x_C \cos\theta + h \sin\theta$$

$$\sin\theta = \frac{H}{l} \qquad \cos\theta = \frac{\sqrt{l^2 - H^2}}{l}$$



F_1 and F_2 can be read from weighbridge

$$z_C = r + \frac{F_2 - F_1}{P} \cdot \frac{1}{H} \cdot \sqrt{l^2 - H^2}$$



End of chapter 3